

Customer Churn Prediction Using Machine Learning

Executive Summary

Customer churn is a critical challenge for telecommunication companies, as losing existing customers directly impacts revenue and profitability. The objective of this project is to develop a machine learning solution capable of predicting customer churn and identifying the key factors influencing customer retention.

Using the Telco Customer Churn dataset, multiple machine learning models were developed and evaluated. Logistic Regression emerged as the best-performing model, achieving an accuracy of 82.04% and a ROC-AUC score of 0.8620. The model provides actionable insights that can help organizations proactively identify at-risk customers and implement targeted retention strategies.

Business Problem

Customer acquisition is significantly more expensive than customer retention. Telecom companies experience customer attrition due to pricing concerns, service dissatisfaction, contract flexibility, and competitive offerings.

The business objective is to predict whether a customer is likely to churn and identify the primary drivers behind churn behavior.

Project Objectives

- Analyze customer demographics and service usage patterns.
- Identify factors contributing to customer churn.
- Build predictive machine learning models.
- Compare model performance.
- Generate actionable business recommendations.
- Develop a deployable churn prediction solution.

Dataset Overview

Dataset: Telco Customer Churn Dataset

The dataset contains customer demographic information, service subscriptions, billing details, contract information, and churn status.

Key Features:

- Gender
- Senior Citizen Status
- Partner
- Dependents

- Tenure
- Internet Service
- Online Security
- Tech Support
- Streaming Services
- Contract Type
- Payment Method
- Monthly Charges
- Total Charges

Target Variable:

- Churn (Yes/No)

Methodology

The project followed a standard machine learning lifecycle:

1. Data Collection
2. Data Cleaning
3. Exploratory Data Analysis (EDA)
4. Feature Engineering
5. Data Encoding
6. Train-Test Split
7. Model Training
8. Model Evaluation
9. Model Selection
10. Model Deployment

Models Evaluated

Logistic Regression

A baseline linear classification model used to estimate churn probability.

Random Forest

An ensemble learning algorithm using multiple decision trees.

XGBoost

A gradient boosting algorithm designed for high-performance predictive modeling.

Model Evaluation Results

Model	Accuracy	ROC-AUC
Logistic Regression	82.04%	0.8620
Random Forest	79.13%	0.8402
XGBoost	81.41%	0.8610

Logistic Regression achieved the highest overall performance and was selected as the final model.

Key Findings

The analysis identified several important churn drivers:

- Customer tenure is the strongest predictor of retention.
- Two-year contracts significantly reduce churn risk.
- Fiber Optic customers demonstrate higher churn propensity.
- Electronic Check users are more likely to churn.
- Online Security and Tech Support services improve customer retention.

Business Recommendations

1. Promote long-term contract plans.
2. Focus retention efforts on newly acquired customers.
3. Encourage automatic payment methods.
4. Increase adoption of value-added services such as Online Security and Tech Support.
5. Implement proactive retention campaigns for high-risk customers.

Deployment Strategy

The selected Logistic Regression model was saved using Joblib and integrated into a Streamlit web application. The application allows users to input customer attributes and receive churn predictions in real time.

Conclusion

The project successfully developed a customer churn prediction system capable of identifying customers at risk of leaving the company. The solution provides both predictive capability and actionable business insights that can support customer retention initiatives and improve long-term profitability.

